

ML4EDF: Physics-Informed Learning of Next-Generation Relativistic Nuclear Energy Density Functionals: From Nuclei to Neutron Stars

Vision: learn the minimal physics missing from current relativistic EDFs and turn it into reliable predictions from finite nuclei to neutron stars.

1. Excellence

1.1 Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)

How can the same theory explain both the binding of finite nuclei and matter compressed inside neutron stars—and how can its predictions be trusted beyond current experimental reach? This is a defining challenge of modern nuclear physics. The NuPECC Long Range Plan 2024 places a predictive understanding of nuclei, reactions and strongly interacting matter at the centre of Europe's strategy [1]. New rare-isotope measurements are reaching increasingly neutron-rich systems, while gravitational-wave and X-ray observations have turned neutron stars into laboratories for dense matter [29]. These advances provide complementary evidence, but also expose a common bottleneck: nuclear models must extrapolate far beyond the data used to construct them. ML4EDF addresses this challenge through one central question: which minimal changes to present relativistic nuclear energy density functionals are jointly demanded by finite nuclei, collective response and neutron-rich matter, and which apparent improvements arise only from the chosen functional representation? The project will answer this by combining paired density-dependent point-coupling (DD-PC) and meson-exchange (DD-ME) calculations with emulation, Bayesian inference and sparse functional correction. A focused quantum-kernel benchmark will test an emerging computational opportunity on the same reduced discrepancy problem, without making the project dependent on quantum advantage.

The scientific stakes extend from the limits of nuclear binding to the properties of neutron-rich matter and neutron stars. Nuclear binding energies and one- and two-nucleon separation energies constrain shell evolution, pairing and the location of particle-emission thresholds, while charge radii probe the spatial distribution of protons and its evolution along isotopic chains. Neutron-skin thicknesses and electric dipole polarizabilities provide complementary constraints on the density dependence of the symmetry energy, particularly around and below nuclear saturation density. The isoscalar giant monopole resonance constrains the incompressibility of finite nuclei and, through energy-density-functional calculations, the incompressibility of symmetric nuclear matter, whereas dipole response provides additional information on the isovector sector. Microscopic calculations of neutron-rich matter and neutron-star mass-radius and tidal-deformability constraints extend these tests towards much larger neutron-proton asymmetries and higher densities. No single observable can identify deficiencies in an energy density functional: progress requires combining complementary observables while separating parameter uncertainty, model-form deficiency and numerical or statistical error. Nuclear density functional theory is particularly well suited to this programme because it provides a self-consistent and computationally tractable framework for describing ground-state and collective properties across medium-mass and heavy nuclei and for connecting them with nuclear-matter predictions [2,3].

The exact nuclear energy density functional (EDF) is unknown. Existing Skyrme, Gogny and relativistic functionals adopt physically motivated analytic forms whose parameters are calibrated to selected data. Better optimisation has produced major advances, yet predictions still separate dramatically toward the neutron drip line and at neutron-star densities. The reason is deeper than uncertain parameter values: the functional form itself may be too restrictive, and an improved fit can hide rather than cure missing physics. Ground-state observables may absorb effects of correlations that appear explicitly in collective motion. Recent configuration-interaction time-dependent relativistic DFT, for example, broadens giant-monopole strength through configuration mixing while leaving the main peak comparatively stable [8]. This illustrates why functional development must use observables with different sensitivities and distinguish robust integral information from correlation-sensitive spectral details. A lower existing functional parameter fitting error alone is therefore not a sufficient measure of progress.

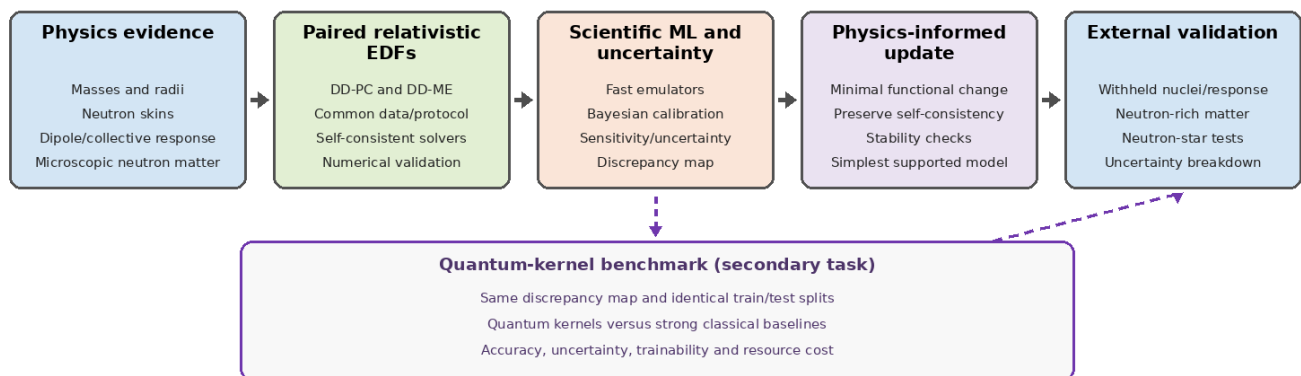
Relativistic nuclear EDFs—the covariant formulation of nuclear DFT—provide an especially powerful test bed. Lorentz symmetry generates the strong spin-orbit interaction naturally and connects finite nuclei, collective response and nuclear matter within the same theoretical language [3]. Density-dependent point-coupling (DD-PC) and meson-exchange (DD-ME) functionals encode this physics in two complementary ways: one uses local contact interactions, while the other represents the interaction through meson fields [4-7]. ML4EDF will use well-tested DD-PC and DD-ME functionals as starting anchors, but it will not interpret their present differences as direct physical effects because they were calibrated with different protocols. Instead, both families will be confronted with the same data, assumptions and validation tests. Agreement after this matched analysis will identify robust trends; persistent differences will quantify the uncertainty associated with choosing a point-coupling or meson-exchange representation.

Scientific machine learning has substantially expanded the statistical analyses that are computationally feasible in nuclear density functional theory. Reduced-order and surrogate models can reproduce selected outputs of expensive nuclear solvers at a fraction of the original computational cost, making repeated evaluations for Bayesian calibration and uncertainty quantification practical [15,18]. Automatic differentiation provides accurate derivatives of EDF observables with respect to model parameters, opening the way to derivative-based optimisation and sensitivity analysis [19]. Machine learning is also moving beyond post-hoc corrections of calculated observables towards learning components or representations of the energy density functional itself. Nuclear studies have constructed orbital-free density functionals trained to reproduce Kohn-Sham or Skyrme calculations [12,13], while a recent proof-of-concept study developed a neural representation of a relativistic point-coupling energy density functional trained on calculations with PC-PK1 [14]. NucleiML [30] and recent emulator-assisted EDF inference [17] further demonstrate that finite-nucleus observables can be incorporated together with nuclear-matter and neutron-star constraints in computationally tractable Bayesian analyses. At the same time, recent work on nuclear binding energies highlights the need to quantify the epistemic uncertainty of ML predictions, particularly when extrapolating away from the training domain [31]. Existing studies therefore establish the individual ingredients—emulation, functional learning, Bayesian inference and ML uncertainty quantification—but largely address them separately. ML4EDF will integrate these capabilities within a matched DD-PC/DD-ME framework, combining ground-state and collective-response information with interpretable functional updates and an explicit separation of experimental, numerical/emulator, parameter and model-form uncertainties.

Quantum computing provides a complementary long-term opportunity. Nuclear calculations for oxygen, calcium and nickel isotopes have reached trapped-ion hardware, and resolution-refinement methods offer a route from tractable low-resolution states to higher-fidelity many-body spaces [24,25]. These advances do not mean that quantum computers can replace global relativistic EDF solvers today. They do, however, motivate rigorous tests of quantum machine learning (QML) on the small, structured datasets produced by expensive nuclear calculations. Quantum kernels can map a compact set of discrepancy coordinates into a high-dimensional feature space without training a deep quantum circuit. The open question is not whether a quantum model can be made to fit, but whether it adds predictive or structural information beyond strong classical kernels under identical data splits and honest resource accounting [26,28]. ML4EDF connects this emerging opportunity directly to its classical workflow: the QML study will interrogate the same low-dimensional map of EDF deficiencies, and no central deliverable will depend on a quantum advantage.

Central hypothesis. ML4EDF tests the hypothesis that the dominant, reproducible deficiencies of current relativistic EDFs can be traced to a limited set of physically interpretable interaction channels and density regimes, rather than to diffuse parameter miscalibration. A candidate modification will be considered physically credible only if it is independently supported by ground-state, collective-response and nuclear-matter observables, remains robust across matched DD-PC and DD-ME formulations, and improves predictions for nuclei and observables excluded from calibration. This cross-observable, cross-representation and out-of-sample consistency provides the key discriminator between a genuine model-form deficiency and an apparent improvement produced by overfitting, compensating errors or omitted correlations. ML4EDF therefore aims not merely to improve fit quality, but to identify **which modifications of the functional are physically warranted and predictive**. This principle is implemented through the evidence-to-prediction workflow of Fig. 1.

Figure 1. ML4EDF evidence-to-prediction workflow



State of the art and project position. Four research fronts have advanced rapidly, but they remain weakly connected. ML4EDF is designed around the missing integration rather than around any single algorithm:

Research frontier	Established advance	Gap addressed by ML4EDF
Relativistic EDFs	Global ground-state descriptions, self-consistent collective response and beyond-mean-field extensions are well established [3–8].	DD-PC and DD-ME use fixed functional forms and different calibration protocols. ML4EDF will place both on a matched footing and test them against the same ground-state, response and nuclear-matter constraints.
ML and functional learning	ML now spans residual corrections, learned nuclear energy densities, neural EDF representations and fast finite-nucleus emulators [11–14,27,30,31].	Most studies correct outputs, emulate fixed solvers or learn from reference models. ML4EDF will diagnose model deficiencies and introduce only sparse, interpretable functional modifications tested self-consistently.
Emulation and uncertainty quantification (UQ)	Reduced-order, Gaussian-process (GP) and empirical-interpolation emulators, automatic differentiation and PMMs increasingly enable large-scale inference [15–19,22].	Ground-state and response surrogates are rarely combined, and uncertainty sources are not always separated. ML4EDF will distinguish experimental, emulator, parameter, model-form and DD-PC/DD-ME representation uncertainties.
Quantum computing / QML	Real-hardware nuclear calculations, state-preparation methods and quantum/quantum-inspired nuclear applications are emerging [24–26,28].	Their role in relativistic-EDF development is still unclear, with no established advantage for this task. ML4EDF will benchmark quantum kernels against strong classical methods on identical data and validation splits.

Ambition beyond the state of the art. ML4EDF advances nuclear-EDF development through an integrated scientific decision framework that connects diagnosis, functional improvement and predictive validation. Collective-response observables contribute directly to identifying where the functional requires additional flexibility; any modification is introduced as a compact, differentiable and physically controlled update rather than a black-box replacement. Calibrating DD-PC and DD-ME under identical statistical assumptions will reveal which inferred trends are robust across the two relativistic representations, while the QML study addresses the same reduced discrepancy problem and validation criteria as competitive classical methods. Scientific success is therefore defined by evidence: either a minimal functional update that improves withheld predictions while remaining consistent with constraints from finite nuclei and neutron-rich matter, or a quantitative demonstration that the existing functional structure is already supported at the resolution of the available data. Both outcomes strengthen the reliability of relativistic-EDF extrapolations and establish a transferable methodology for future global functional development.

Expected scientific insight. ML4EDF will disentangle three closely connected questions in global EDF development. First, how far can the existing parameterisation account for the combined ground-state, collective-response and matter evidence once all major uncertainty sources are treated consistently? Second, which physical conclusions remain robust when DD-PC and DD-ME are calibrated and validated under the same conditions? Third, where a persistent discrepancy is supported by both representations and multiple observable classes, what is the smallest physically admissible modification that improves the affected sector while preserving successful predictions elsewhere? The resulting information map will identify which nuclei and observables constrain each sector of the relativistic EDF, where additional information would most reduce uncertainty, and which future measurements would provide the greatest discriminating power. The QML benchmark adds a forward-looking fourth question: whether quantum feature spaces provide complementary predictive or structural information beyond strong classical baselines for this reduced nuclear-physics problem.

Research objectives. Four measurable objectives translate the central hypothesis into verifiable scientific decisions:

Objective	Research target and verifiable criterion
O1. Validated relativistic-EDF emulators	Build and validate fast emulators for selected DD-PC/DD-ME ground-state and collective-response observables. Success requires agreement with held-out full-solver calculations within observable-specific tolerances and calibrated predictive uncertainty.
O2. Multi-observable diagnosis	Determine which EDF sectors are constrained by ground-state, collective-response and microscopic-matter evidence, and identify where additional functional flexibility is supported. Posterior-predictive, sensitivity and cross-observable consistency tests will quantify each conclusion.

Objective	Research target and verifiable criterion
O3. Minimal functional update	Test compact, differentiable modifications of selected density-dependent interaction channels. An update is retained only when it improves grouped external validation while satisfying self-consistency, stability and nuclear-matter constraints; otherwise the validated baseline is retained.
O4. Extrapolation and quantum diagnostic	Validate predictions for withheld nuclei and observables, propagate validated models to neutron-star constraints, and benchmark quantum against classical kernels on the same discrepancy coordinates. Performance will be assessed through accuracy, uncertainty, robustness and computational resource requirements.

1.2 Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices)

Overall design and scope. ML4EDF follows a four-stage evidence-to-prediction logic: establish a common high-fidelity baseline, emulate and diagnose, introduce evidence-supported functional updates, and validate the resulting predictions. Method1–Method2 deliver the validated emulators of **O1**; Method2–Method3 provide the multi-observable diagnosis of **O2**; Method3 implements the minimal functional update of **O3**; and Method4 delivers the external and quantum diagnostics of **O4**. DD-PC and DD-ME will be placed on a common numerical and statistical footing so that robust physical trends can be distinguished from representation dependence. The benchmark set will emphasise selected closed-shell and semi-magic O, Ca, Ni, Zr, Sn and Pb nuclei. Binding and separation energies constrain bulk, shell and pairing trends; charge radii probe proton-density and surface systematics; neutron skins and dipole polarizabilities constrain the isovector density dependence; selected monopole and dipole observables probe incompressibility and isovector dynamics; and microscopic neutron matter constrains neutron-rich matter. Full response distributions and neutron-star mass-radius and tidal-deformability observables provide stringent external tests. This design builds directly on the researcher’s work connecting dipole polarizability with neutron-star structure and low-energy dipole strength with astrophysical reactions [9,10].

Method1 – Reproducible high-fidelity baseline and data design (O1). A common computational protocol will be established for DD-PC and DD-ME, covering physical constants, pairing, basis truncation, convergence and observable definitions. Each calculation will retain model parameters, code version and convergence diagnostics, and selected reference cases will be independently reproduced before training data are generated. Parameter domains will be centred on established parametrisations and restricted by physical and numerical admissibility. Space-filling designs will initialise the training set, followed by active learning in regions of large emulator uncertainty or posterior sensitivity. Parameter-space, isotopic-chain and observable-class holdouts will separate interpolation, extrapolation and transfer between ground-state and response information.

Method2 – Emulation, sensitivity and uncertainty quantification (O1–O2). Gaussian processes will provide transparent emulators for scalar observables, while reduced-order representations and parametric matrix models will be tested for collective response, where retaining spectral information is advantageous [20]. Existing reduced-basis, empirical-interpolation and PMM studies demonstrate the feasibility of accurate nuclear emulation [15,18,22]. Emulator uncertainty will be propagated explicitly, with cross-validation, coverage tests and targeted full-solver checks defining each surrogate’s validated domain. Sensitivity and information-gain analyses will then identify which observables constrain each EDF sector and where additional data would most reduce uncertainty, providing the quantitative basis for **O2**.

Method3 – Matched Bayesian diagnosis and minimal functional update (O2–O3). DD-PC and DD-ME will be analysed using the same evidence, likelihood structure, matched physical prior principles and validation strategy. The hierarchical model will distinguish measurement, numerical/emulator, parameter and residual model-form uncertainties. Additional flexibility will be considered only where a coherent discrepancy is supported by multiple observable classes and persists across both relativistic representations. ML4EDF will then test a compact set of smooth, physically interpretable modifications to selected density-dependent coupling channels rather than replacing the EDF with an unrestricted neural model. Each candidate will preserve the corresponding rearrangement and residual-interaction contributions required for self-consistency. Sparse Bayesian selection and grouped external validation will favour the simplest supported update, which will then be tested with the complete solvers for convergence, saturation, stability and causal behaviour over the relevant density range. Symbolic regression, if used, will only summarise an already stable inferred trend in compact analytic form.

Method4 – External validation and connected QML benchmark (O4). Posterior predictions will be tested on nuclei, isotopic chains and observable classes excluded from calibration. The resulting uncertainty atlas will separate experimental, numerical/emulator, parameter and model-form contributions, while DD-PC/DD-ME representation dependence will be reported explicitly. Validated results will be propagated to symmetry-energy

indicators, neutron skins, dipole response and neutron-star mass-radius and tidal-deformability relations, with the predictive domain defined by full-solver convergence and physical-admissibility criteria.

The QML benchmark will use the same low-dimensional discrepancy coordinates identified in **O2–O3**. Quantum-kernel regression and classification will be compared with Gaussian-process (GP), radial-basis and classical support-vector methods using identical grouped data partitions. Performance will be assessed through predictive accuracy, uncertainty calibration, robustness to noise and computational resource requirements. Simulation will provide the primary controlled comparison, with a focused hardware demonstration only where it adds scientifically interpretable evidence. The benchmark will establish where quantum feature spaces provide complementary predictive or structural information and where classical approaches remain preferable.

Decision gates and validation criteria. The staged design links each methodological advance to explicit evidence, ensuring that conclusions are supported by validated emulators, independent tests and physically admissible parameter regions:

Gate	Evidence and decision
G1 (Month 8)	Advance to Bayesian inference for observables and parameter regions that meet pre-defined emulator accuracy and uncertainty-coverage criteria.
G2 (Month 11)	Introduce additional functional flexibility only where matched DD-PC/DD-ME analyses and independent observable classes support a coherent, physically interpretable discrepancy.
G3 (Month 16)	Select the simplest functional update that improves grouped external validation while satisfying self-consistency, stability and nuclear-matter constraints; otherwise retain the validated baseline.
G4 (Month 20)	Advance to extrapolative predictions after successful tests on withheld isotopic chains and observable classes, together with neutron-matter and neutron-star consistency checks.
G5 (Month 22)	Identify added value from QML only where quantum kernels provide predictive or structural information beyond competitive classical baselines under comparable resource requirements.

Interdisciplinarity and trustworthy AI. Nuclear theory defines physically admissible model changes and validation constraints; Bayesian statistics integrates heterogeneous evidence and separates uncertainty sources; scientific ML enables emulation, sensitivity analysis and structured model discovery; and quantum information provides a complementary representation for the final discrepancy problem. Their integration ensures that data-driven conclusions remain transparent and physics constrained. Fixed data partitions, competitive classical baselines, documented preprocessing and hyperparameters, uncertainty quantification (UQ), and validation with the full self-consistent solvers will accompany all ML/QML results, with the validated domain stated explicitly. Interpretability will be assessed through reproducible sensitivity patterns, robustness under resampling, consistency across DD-PC and DD-ME, and the physical behaviour of the resulting functional. AI-based tools remain under human scientific oversight and are used exclusively for scientific modelling and inference.

Gender dimension and open science. The gender dimension is not relevant to the research content because ML4EDF concerns atomic nuclei, nuclear matter and computational methods and involves no human participants or personal or biological data. Inclusive practices will nevertheless guide mentoring, training and public engagement. A Data Management Plan will be delivered by **Month 6**. Version-controlled software, automated tests, metadata and reproducible workflows will support transparency and reuse. Shareable derived data, posterior samples and validated software releases will be deposited in appropriate repositories with persistent identifiers. Publications will be open access and preprints shared early, following the principle “**as open as possible, as closed as necessary.**”

1.3 Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host

Supervision. Prof. Morten Hjorth-Jensen combines nuclear many-body theory with current research in Bayesian and physics-informed ML, parametric matrix models, neural-network quantum states, quantum algorithms and QML. His recent work includes interpretable PMMs [20], Bayesian extrapolation of coupled-cluster nuclear matter [21] and quantum state preparation by resolution refinement [25]. This expertise is directly complementary to the researcher’s relativistic EDF, collective-response and nuclear-astrophysics background. Weekly scientific meetings will address physical interpretation, algorithms and software; monthly progress reviews will track milestones and risks. A Career Development Plan will be agreed in Month 2 and reviewed every six months. The researcher will lead solver integration, data generation, inference and publications, while the supervisor will guide Bayesian validation, PMM/QML methodology, links to many-body theory and international network development.

Personalised training. Training is tied to project tasks and to the transition from an application-oriented nuclear theorist to an independent developer of uncertainty-aware nuclear models:

Training area	Mechanism and project use	Measurable evidence
Scientific ML and UQ	Supervisor/group tutorials and selected advanced-ML training; GPs, PMMs, active learning and hierarchical Bayes applied in WP1–WP3.	Validated emulators, calibrated posteriors and lead-author methodology output.
Many-body theory	Seminars and discussions connecting relativistic EDFs with microscopic nuclear-matter calculations, collective response and reduced-order modelling.	Physically motivated priors, validation criteria and nuclear-matter constraints used in the analysis.
Quantum computing /QML	Relevant quantum-computing/QML training and a supervised benchmark task integrated into WP3.	Reproducible quantum/classical benchmark with accuracy, uncertainty, robustness and resource metrics.
Open software and leadership	Open-science and research-data-management training; code review, mentoring, project management, grant writing and scientific communication.	FAIR software/data release, documented mentoring and seminars, Career Development Plan, and a follow-on proposal concept.

Two-way knowledge transfer. From host to researcher: Bayesian scientific ML, PMMs, active learning, uncertainty calibration, neural representations of correlated systems, quantum state preparation and critical QML benchmarking. From researcher to host: DD-PC/DD-ME relativistic EDFs, relativistic Hartree-(Bogoliubov) and QRPA/finite-temperature QRPA implementations, E1/M1 and compressional response, dipole-polarizability/neutron-star correlations, global mass modelling, ensemble learning and quantum-kernel residual prediction. Transfer will occur through internal discussions, seminars, documented codes, shared software, mentoring of a student task and co-authored publications. The exchange is genuinely complementary: UiO contributes advanced many-body, statistical and quantum methodology, while the researcher brings a mature relativistic EDF platform and response expertise on which those methods can generate new physics.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

Dr Tanmoy Ghosh obtained his PhD in 2024 and is a postdoctoral researcher at the University of Zagreb. His research connects nuclear structure, reactions and astrophysics: shell-model level densities and reaction inputs; the density dependence of symmetry energy in nuclear reactions [32]; relativistic EDFs and relativistic/finite-temperature QRPA; collective E1/M1 response; neutron skins; dipole polarizability; and neutron-star observables. Recent publications establish relations between dipole polarizability and neutron-star structure [9] and show that astrophysical reaction-rate enhancements depend on the alignment of low-energy dipole strength with the neutron threshold [10]. This combination of ground-state, response and astrophysical expertise is essential for selecting physically meaningful learning targets and designing validation tests that go beyond mass fitting.

In parallel, he has developed an ML/QML line for nuclear masses. A physics-guided ensemble and model-averaging framework reports sub-100-keV benchmark accuracy and systematic external validation [27], while his quantum-kernel study provides practical experience with feature maps, fidelity kernels, overfitting and simulator-based assessment [28]. He is proficient in Python, C++, Fortran, ROOT, Qiskit and high-performance workflows. ML4EDF is therefore a coherent independence step: from correcting outputs of fixed models to leading physics-informed construction and uncertainty analysis of the functional itself.

2. Impact

2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his skills development

Career transformation. The long-term objective is to lead an independent programme at the interface of nuclear many-body theory, nuclear astrophysics, scientific ML, uncertainty quantification (UQ) and quantum computing. The researcher brings established expertise in relativistic EDF/QRPA, nuclear-structure and nuclear astrophysics, together with practical experience in ML and quantum computing. The fellowship will deepen this foundation through advanced Bayesian inference, interpretable scientific ML, reduced-order modelling, research-software leadership and rigorous QML benchmarking. By integrating the full chain from physical modelling and high-fidelity solvers to inference, validation and open software, ML4EDF will establish a distinctive research profile in **trustworthy data-driven nuclear theory**.

Development need	Action and measurable evidence
Bayesian inference and UQ	Develop hierarchical inference and uncertainty-aware functional decisions; demonstrated through validated posterior products and a lead-author methodology output.
Research-software leadership	Build a tested, reusable relativistic-EDF/ML workflow with DOI, documentation, tests and reproducible examples.

Development need	Action and measurable evidence
Advanced quantum/QML expertise	Extend existing quantum-computing experience to physics-informed quantum kernels and controlled quantum/classical benchmarks.
Leadership and funding	Mentor a student task, deliver seminars/tutorials and develop an independent follow-on ERC proposal, with progress tracked through the Career Development Plan (CDP).

These skills, outputs and networks will strengthen independence and employability across nuclear theory, computational science and quantum/AI research. Progress will be demonstrated through lead-author outputs, software ownership, mentoring, scientific presentations and an independent ERC funding concept. UiO's Faculty of Mathematics and Natural Sciences additionally offers successful European Fellowship researchers an **internally funded third year of employment**, outside the 24-month MSCA project. This continuity will allow the researcher to consolidate ML4EDF results, develop follow-up directions, strengthen teaching and supervision experience, and prepare competitive independent funding applications.

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities

Dissemination, exploitation and communication target distinct audiences and complementary forms of impact. Research outputs will be accompanied by sufficient methodological, validation and uncertainty information to support scrutiny, reproducibility and reuse.

Activity	Audience and measures	Timing / indicators
Dissemination	Nuclear-structure, astrophysics, scientific-ML and quantum communities: three open-access manuscripts, at least three international conference contributions, two seminars, preprints and reproducible benchmark notebooks.	Months 7–24 ; manuscripts, presentations, repository downloads and scientific uptake.
Exploitation	EDF developers, computational nuclear theorists and research-software users: documented emulator/calibration workflow, compact functional-update forms, validation suite and reusable examples under a clear license.	Prototype Month 8 ; releases Months 16 and 24 ; downloads, issues/forks and reuse requests.
Communication	Students and science-interested public: project webpage, two accessible visual stories and one UiO engagement activity illustrating trustworthy AI and quantum methods in nuclear science.	Launch Month 3 ; updates every six months; attendance and online engagement.

Intellectual property will be assessed before public release. Open dissemination and licensing will be the default where appropriate, as the main exploitable outputs are software, workflows and compact functional-update forms. Where protectable value emerges, UiO innovation support will be consulted before disclosure. Authorship, software contributions and data ownership will be agreed at project start and reviewed at key milestones.

2.3 The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Impact pathway	Expected contribution and credible magnitude
Scientific	Response-aware, uncertainty-resolved diagnosis of relativistic EDF structure; either a compact validated functional update or a well-supported baseline; matched DD-PC/DD-ME analysis, three papers and an open uncertainty atlas.
Methodological / technological	Reusable emulation, active-learning and Bayesian workflows that reduce the cost of repeated EDF/QRPA inference; quantified acceleration and accuracy, plus a reproducible quantum-kernel benchmark defining where QML adds value.
Societal and European capacity	More reliable nuclear theory supports interpretation of rare-isotope and multimessenger data, while the project develops interdisciplinary expertise in nuclear theory, trustworthy scientific ML, open software and quantum methods. Impact will be monitored through uptake, collaborations and follow-on funding.

The immediate impact of ML4EDF is primarily scientific and methodological. It will determine whether persistent discrepancies in relativistic EDF predictions arise from parameter uncertainty, functional-form limitations, numerical/emulator effects or representation dependence. The uncertainty atlas and information-gain analysis will identify which observables most strongly constrain each EDF sector and which future measurements could most effectively reduce uncertainty. Matched DD-PC/DD-ME analysis will provide a controlled measure of representation dependence, while open emulators and validation workflows will enable efficient recalibration and extension to additional nuclei. In the longer term, improved predictions for neutron-rich nuclei and dense matter

will strengthen the interpretation of rare-isotope and multimessenger observations and contribute to the NuPECC goal of increasingly predictive nuclear theory.

The project is designed so that every scientifically supported outcome remains reusable. Where the existing functional structure is sufficient at the resolution of the available evidence, the validated baseline and uncertainty decomposition will define its reliable predictive domain and highlight where additional observables or extended many-body treatments would be most informative. Where a functional update is supported, its compact form and matched validation will make it directly testable and extendable by other groups. For experimental programmes, information-gain analysis will prioritise measurements by their expected constraining power. For the ML and quantum communities, ML4EDF provides a demanding physics benchmark in which predictive accuracy, uncertainty, extrapolation and physical consistency are assessed together. This combination strengthens the credibility, reuse and transferability of the project outcomes beyond the initial nuclei and EDF formulations.

3. Quality and Efficiency of the Implementation

3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

Work plan and implementation logic. The work plan translates the four objectives and methodological steps into four interconnected Work Packages (WPs). **WP1** implements Method 1 and the emulator component of Method 2 to deliver the validated computational framework of **O1**. **WP2** completes Method 2 and implements Method 3, providing the multi-observable diagnosis of **O2** and the evidence-based functional update of **O3**. **WP3** implements Method 4, delivering the external validation, astrophysical propagation and QML diagnostic of **O4**. **WP4** supports all objectives through continuous training, management, open science and impact activities. The sequence follows the evidence-to-decision logic of Fig. 1: validated emulators precede inference, functional updates follow matched baseline diagnosis, and extrapolative conclusions are supported by independent validation.

WP / timing / effort	Main tasks	Deliverables and milestones
WP1 Data and emulators / M1–M8 / 30%	Common DD-PC/DD-ME protocol; curated evidence; parameter-space design; Gaussian-process (GP) and structure-aware emulators; active-learning validation.	D1.1 workflow/data specification (M4); D1.2 validated emulators (M8); Milestone 1: G1 passed.
WP2 Diagnosis and functional update / M5–M16 / 35%	Matched Bayesian inference; sensitivity and uncertainty quantification; evidence-supported functional updates; grouped validation and physical filtering.	D2.1 baseline posterior and uncertainty report (M11); Milestone 2: G2 passed; D2.2 functional-update decision and validated candidate(s), or retained baseline (M16); Milestone 3: G3 passed.
WP3 External validation and QML / M13–M22 / 25%	Withheld-chain and response tests; representation dependence; neutron-star propagation; classical/quantum kernel benchmark.	D3.1 validation and uncertainty atlas (M20); Milestone 4: G4 passed; D3.2 QML benchmark (M22); Milestone 5: G5 completed.
WP4 Training, management and impact / M1–M24 / 10%	Career Development Plan (CDP), supervision, Data Management Plan, code quality, publications, dissemination, exploitation and communication.	Career Development Plan (M2); Data Management Plan (M6); mid-term review (M12); final open release and communication package (M24).

Feasibility and resource logic. ML4EDF builds on existing DD-PC/DD-ME fitting and response codes, the researcher’s experience in their use and development, and a benchmark set focused on nuclei with robust self-consistent solutions. Computationally intensive response calculations will be prioritised through active learning and emulated only within validated domains. The host contributes advanced Bayesian, reduced-order and quantum methodology, while UiO computing and software infrastructure supports reproducible implementation and open release. Effort is concentrated in WP1–WP2 because the computational baseline, emulators and multi-observable diagnosis underpin all subsequent outputs. Functional-update and QML activities are governed by explicit decision gates, ensuring that the core scientific outputs remain independent of additional functional flexibility or quantum performance.

Gantt chart. Dependencies place functional-update decisions after emulator validation and extrapolative conclusions after independent testing. **M denotes project month.**

Activity	M1-3	M4-6	M7-9	M10-12	M13-15	M16-18	M19-21	M22-24
WP1 Data	■	■	■					
WP2 Diagnosis		■	■	■	■	■		
WP3 Validation					■	■	■	■
WP4 Training	■	■	■	■	■	■	■	■
Key outputs	CDP	D1.1/ Data Plan	D1.2/ G1	D2.1 / G2 / review		D2.2 / G3	D3.1 / G4	D3.2 / G5 / release

Risk management. Risks will be reviewed monthly and at each milestone, with contingencies designed to preserve the principal scientific outputs:

Risk (likelihood / impact)	Mitigation	Contingency
Emulator accuracy limited in sensitive regions (medium/high)	Active learning, transformed outputs, GP and structure-aware alternatives; explicit propagation of emulator uncertainty.	Restrict inference to validated domains and use targeted full-solver calculations for posterior verification.
Reduced solver convergence at extreme parameters (medium/medium)	Physics-informed priors, staged domain expansion and automated convergence diagnostics.	Restrict predictions to physically and numerically validated regions.
Functional update lacks robust support (medium/high)	Sparse priors, grouped holdouts, self-consistent rearrangement/residual terms and stability/causality filters.	Reduce the update to the simplest supported form or retain the validated baseline.
Response calculations become computationally demanding (medium/low)	Prioritise robust moments/centroids and benchmark nuclei; develop response emulators early.	Reduce response calibration while retaining external response validation.
Limited added value from quantum kernels (high/low)	Strong classical baselines, simulator-first design and resource diagnostics.	Publish the planned boundary-of-use result; no core outcome depends on quantum advantage.

Management and quality control. The researcher leads all WPs under the supervisor's methodological and strategic guidance. Weekly meetings, monthly risk reviews and six-monthly Career Development Plan updates will monitor progress and scope. Deliverables will undergo reproducibility, uncertainty-calibration, full-solver validation and documentation checks, while Git-based testing and versioned computational environments will preserve provenance.

3.2 Quality and capacity of the host institutions and participating organisations, including hosting arrangements

The University of Oslo (UiO), Faculty of Mathematics and Natural Sciences, provides a strong environment for ML4EDF at the intersection of nuclear many-body theory, scientific ML and quantum computing. The researcher will be integrated into Prof. Hjorth-Jensen's nuclear many-body and computational-physics activities through weekly supervision, group meetings, seminars, relevant graduate training and access to institutional and national high-performance computing. The host contributes expertise in quantum many-body methods, Bayesian ML, parametric matrix models, neural-network quantum states, quantum state preparation and QML [20–25], complementing the researcher's specialisation in relativistic EDFs, DD-PC/DD-ME and QRPA. Conversely, the researcher brings a self-consistent relativistic EDF/QRPA platform to UiO's data-driven and quantum-method activities. This complementarity makes two-way knowledge transfer an integral part of the research.

Hosting will begin with induction in computing, research integrity, data management and administrative procedures. UiO will provide office and computing facilities, software/repository and open-science support, library resources, mobility assistance and EU-project administration. The supervisor will facilitate integration into relevant many-body, ML and quantum networks and support conference strategy, collaborations and independent proposal development. Combining the researcher's established relativistic EDF/QRPA tools with UiO's computational expertise will enable rapid project start-up and sustained methodological exchange. For successful European Fellowship holders, the Faculty also provides an **internally funded third year of employment**, outside the 24-month MSCA project, offering continuity to consolidate results, strengthen teaching and supervision experience, develop follow-up research and prepare independent funding applications.

Hosting arrangements mapped to project needs.

Project need	UiO provision and direct use
Scientific integration	Weekly supervision, group meetings and seminars; rapid interpretation and milestone decisions.
Methods and computing	Supervisor-led methodological guidance, relevant ML/quantum training, high-performance computing and reproducible software environments; direct support for Bayesian, reduced-order and QML workflows.
Open science and software	Library, research-data and repository support; Data Management Plan, FAIR data/software, testing, licensing and persistent identifiers.
Career, mobility and administration	Career counselling, International Staff Mobility Office and Faculty EU support; Career Development Plan, relocation/reporting support, network development and follow-on funding preparation.

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